Course Syllabus * SPRING 2023

INST 327 – Database Design and Modeling

All Sections

Course Description and Learning Outcomes

A broad introduction to relational database systems, this course will provide students with a combination of conceptual understanding and technical practice. Students will learn about the relational model, which provides the logical framework for designing and querying relational databases. Students will also learn important technical and conceptual approaches to database design, including user-oriented design, requirements analysis and specification, entity relationship modeling, and normalization. Students will put these fundamentals into action by learning and using the Structured Query Language (SQL) and a database management system (DBMS) to build, populate, and query a working database.

After successfully completing the course, you will be able to:

- Describe the relational model as a logical system for structuring data for retrieval.
- Create user-oriented database queries using the Structured Query Language (SQL).
- Translate user needs into functional database requirements by using entity-relationship models that conform to the relational model.
- Build a working relational database using a database management system (DBMS).
- Normalize and denormalize a relational database to optimize performance.
- Identify security issues in databases and develop approaches to address them.
- Explain how the use and design of relational database systems reflect broader social and organizational structures and the related ethical and equity issues.

Instructors

Prof. Pamela Duffy, pronouns: she/her
Iskander Lou, pronouns: he/him

In-Class Meetings

Lecture:

010x	W	2:00-2:50pm	BRB1101
020x	Th	11:00-11:50am	EGR1202
ESG1	F	10:00-10:50am	BLD4 4335

Discussion/Lab Sessions:

0101	F	9:00-9:50am	ATL 2428
0102	F	10:00-10:50am	BPS 1236
0103	F	11:00-11:50am	CCC 1111
0201	F	2:00-2:50pm	EGR 1108
0202	F	3:00-3:50pm	EGR 1108
0203	F	4:00-4:50pm	EGR 1108
ESG1	F	10:50-11:45am	BLD4 4335

Office Hours: Posted on ELMS

Teaching Assistants	Sections
Bingqi Lian (UTA) pronouns: he/him	0101
Kamran Djourshari (Lead GTA) pronouns: he/him	0102
James van Doorn (UTA) pronouns: he/him	0103
Alekya Ghanta (GTA) pronouns: she/her	0201
Srikanth Parvathala (GTA) pronouns: he/him	0202
Nidhi Nambiar (GTA) pronouns: she/her	0203
Pranav Adijaru (GTA) pronouns: he/him	ESG1
Vinya Dengre (GTA)	pronouns: she/her
Hrushi Patel (UTA)	pronouns: he/him
Claire Jacobs (UTA)	pronouns: she/her
Bo Wu (UTA)	pronouns: he/him
Arya Thakur (UTA)	pronouns: she/her

Academic Peer Mentors: ELMS

Prerequisites

Prerequisite: Minimum grade of C- in INST126
Restriction: Must be in the Information Science program.
Credit granted for: INST327 or BMGT402

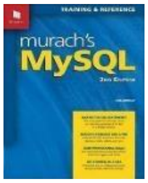
Course Communication

- Use ELMS for messages to instructors, TAs, and AMPs.
- Message subject headings should begin with "**INST327 -** ".
- Project messages should include the team number in the heading.
- Course information will be posted on ELMS. Information will be communicated via the Inbox, Announcements, and Discussion areas. Customize your alerts to ensure you receive notifications via your preferred email account.

Required Resources

Course ELMS Website: **https://umd.instructure.com/courses/1343300**

Textbook: Murach's MySQL, Third Edition (2019)
Murach, J. ISBN: 9781943872367



Physical copies are available via campus bookstore.
eBook: https://www.murach.com/shop/murach-s-mysql-3rd-edition-detail

Additional readings and resources may be assigned via ELMS.

We will work through SQL examples during lecture. Please have a computer available for course-related work.

Software: https://dev.mysql.com/downloads/mysql/

- MySQL Community Server v8 or later (alternative XAMPP, see ELMS)
  - NOTE: For MacOS, specific sub-versions of MySQL are compatible with specific MacOS versions.
- MySQL Workbench v8 or later
  Please check ELMS for download and installation instructions, videos, and other resources. Feel free to seek help from the instruction team and the peer mentors (AMPs).
- Turning Point (PointSolutions) software available via TerpWare will be used to enhance student engagement and track participation in classes. You may use the account login, if you prefer not to load the software on your device. **NOTE:** NO requirement to purchase a physical Clicker device.

Campus Policies

It is our shared responsibility to know and abide by the University of Maryland's policies that relate to all courses, which include topics such as:

- Academic integrity
- Student and instructor conduct
- Accessibility and accommodations
- Attendance and excused absences
- Grades and appeals
- Copyright and intellectual property

Please visit www.ugst.umd.edu/courserelatedpolicies.html for the Office of Undergraduate Studies' full list of campus-wide policies and follow up with me if you have questions.

Course Policies

- Stay engaged with the course and participate.
- There are many ways you can participate, including completing asynchronous work, attending lecture and discussion sessions, discussions, pair or group work, and project teamwork.
- Have a computer available for course-related work and bring it to class; (remember to fully charge it).
- Monitor ELMS daily; adjust settings as needed to ensure receipt of all notifications.
- Follow course calendar activities:
  - Watch the week's assigned videos, keep up with readings, and execute examples before the lecture.
  - Review practice problems before discussion/lab sessions.
  - Complete worksheets, homework, and quizzes as assigned.
  - Complete lab quizzes during discussion sessions (or soon after) and submit via ELMS.
  - Submit all individual and team assignments via ELMS on time (Some assignments cannot be submitted late, and others have late penalties. If you have an excuse, contact the instructors or TA before the deadline whenever possible!).
  - Do not wait until the deadline to make your ELMS submission! ELMS will mark submissions as "late" even if the submission is only late by 1 or 2 minutes! Technical issues at 10 pm on the due date are NOT our emergency! So, begin assignments early and attend office hours for assistance.
- All team members must participate actively in the team project.
- If you are struggling with course content, course schedules, please reach out to the instructors and TAs.

Additional University Resources

You are expected to take personal responsibility for your own learning. This includes acknowledging when your performance does not match your goals and doing something about it. Everyone can benefit from some expert guidance on time management, note taking, and exam preparation, so I encourage you to consider visiting http://ter.ps/learn and schedule an appointment with an academic coach. Sharpen your communication skills by visiting https://english.umd.edu/writing-programs/writing-center and schedule an appointment with the campus Writing Center. Finally, if you just need someone to talk to, visit http://www.counseling.umd.edu.

Notice of Mandatory Reporting

Notice of mandatory reporting of sexual assault, sexual harassment, interpersonal violence, and stalking: As faculty members, we are designated as a "Responsible University Employees," and we must report all disclosures of sexual assault, sexual harassment, interpersonal violence, and stalking to UMD's Title IX Coordinator per University Policy on Sexual Harassment and Other Sexual Misconduct. If you wish to speak with someone confidentially, please contact one of UMD's confidential resources, such as CARE to Stop Violence (located on the Ground Floor of the Health Center) at 301-741-3442 or the Counseling Center (located at the Shoemaker Building) at 301-314-7651. You may also seek assistance or supportive measures from UMD's Title IX Coordinator, Angela Nastase, by calling 301-405-1142, or emailing titleIXcoordinator@umd.edu. To view further information on the above, please visit the Office of Civil Rights and Sexual Misconduct's website at ocrsm.umd.edu.

Names/Pronouns and Self Identifications

The University of Maryland recognizes the importance of a diverse student body, and we are committed to fostering equitable classroom environments. We invite you, if you wish, to tell us how you want to be referred to both in terms of your name and your pronouns (he/him, she/her, they/them, etc.). The pronouns someone indicates are not necessarily indicative of their gender identity. Visit trans.umd.edu to learn more.

Additionally, how you identify in terms of your gender, race, class, sexuality, religion, and dis/ability, among all aspects of your identity, is your choice whether to disclose (e.g., should it come up in classroom conversation about our experiences and perspectives) and should be self-identified, not presumed or imposed. We will do our best to address and refer to all students accordingly, and we ask you to do the same for all your fellow Terps.

Office Hours, Topic Clinics, Group Project Support

For extra clarification of a lecture topic, lab exercise, quiz, or assistance with a homework assignment, please attend online or in-person office hours. In addition to the instruction team's office hours, there will be peer mentor (AMP) office hours that will be announced via ELMS.

Course Structure

The course is delivered in a hybrid learning format structured around weekly modules. As such, there are asynchronous elements and synchronous classes.

- *Self-study Preparation (Asynchronous Materials)*: This is a combination of readings, slides, and videos. Students are expected to complete the assigned readings and videos prior to the weekly lecture class time.

- *Lecture and Discussion Classes (Synchronous Sessions):* Lectures and discussion/lab classes will be interactive and based on the material covered in the *Self-Study Preparation*. Participation is necessary to excel at this course. If you anticipate missing a session, email the instructors and TAs ahead of time, or contact us as soon afterwards as possible. At least three "participation instances" (e.g., lecture + lab + one office hour visit or lecture + lab + meeting with your project team) per week is advised and encouraged.

Course Activities and Assessments

This course involves formative and summative assessment components, as well as an applied team project, which serves as the Scholarship in Practice component for the course. Many of the activities and assessments are designed to support the semester-long Team Database Design project. Activities gauge your understanding and expertise of the SQL skill set and readiness to perform the related project work.

- *Video Quizzes (5% of final grade):* Weekly videos with embedded short multiple choice and T/F questions provide asynchronous preparation for the synchronous class sessions and will check your understanding of the concepts and skills covered in the videos. Students will be expected to watch the videos and complete the video quizzes prior to attending the relevant synchronous class session. [F/I]
- *Discussions (15% of final grade):* Bi-weekly, asynchronous online discussions will help with your understanding of the concepts and skills covered in the course. Students will be expected to post a main contribution, responses to other students, and a final contribution summarizing their take-aways from the exchange. [F/I]
    - *Ethics Discussions:* Biweekly, students will be asked to discuss data ethics readings from the previous two weeks.
    - *Muddiest Point Clarifications:* Periodically after major, complex topics, students will be asked to discuss a concept or skill covered in the previous two weeks that they did not understand well. The initial post may be written, video, slideshow, or similar artifact to clarify the topic/point of their choice.
- *Lab Exercises (10% of final grade):* Lab exercises provide practice of the skills necessary for a comprehensive understanding of SQL and the successful completion of the team project. You will receive these practice problems ahead of time and should preview them before the discussion class, but you must execute them during the discussion class with the TA's or AMP's assistance as needed. You will submit your work individually via ELMS and will receive a completion grade. Lab exercises will take place during your in-person discussion session and many will be done with your project team to foster team cohesion. On discussion/lab days, attendance is required for full credit on the lab exercise. Sign-in logs will be used to track attendance. [F/I]
- *Quizzes (5% of the final grade):* Quizzes will test your comprehension of concepts and skills covered in the preceding module. Quizzes are individual work. Most quizzes will become available during the week and will be due on the following Monday. Quizzes will have a 30-minute time limit from when you begin the quiz. [S/I]
- *Homework Assignments (25% of final grade):* There will be several assignments, each of which will include multiple questions. The questions will be practical tasks, such as writing SQL queries, constructing a view, normalizing a table, or developing a stored procedure. The assignments are individual work. Although you may consult with your classmates, AMPs, TAs, and the instructors to develop general approaches to solving questions, you must work individually while you build, type, test, and debug your answers. Assignment questions will be available on ELMS. Completed assignments will be submitted via ELMS. Timely submission of the completed assignments is essential. The due date of each assignment will be stated explicitly in ELMS. If an assignment due date is a religious holiday for you, please let the instructors know as soon as the assignment is announced so that an alternate due date can be set for you. (See course schedule below for due dates). [S/I]
- *Team Project (35% of the final grade):* Students will work in ~5-person teams to design and build a non-trivial relational database. As a Scholarship in Practice course, project-related work is central to this course. The project will involve defining the team structure and responsibilities for individual team members, identifying an information need for a relational database, determining the requirements for the database, developing a deadline-oriented plan and schedule for building the database, designing the logical specifications, building, and populating the database, and developing queries/views that will showcase the capabilities of the

database for fulfilling the identified user needs. Students will choose their teams or be assigned by the instructors. Teams will choose their topics from a list of possible project topics. Central to this project will be team management and team interaction. Project deliverables throughout the semester will include assessment of team function as well as the software development process. Academic Peer Mentors (AMPs) will act as team mentors throughout the life cycle of the project. [A/T+I]

- *Team Status (5% of the final grade):* Project teams will provide periodic updates about their progress on the project. The updates may be asked to be presented as write-ups, slideshows, or brief recorded videos. Details will be made available during the semester. [A/T]

[F: Formative (30%) - S: Summative (30%) - A: Applied (40%) | I: Individual (62%) - T: Team (38%)]

Grades

Your course grade is determined by your performance on the learning assessments in the course. Assessment scores will be posted on ELMS. If you would like to discuss your grade or have questions about how something was scored, please schedule a time with the course TA. Grade disputes must be submitted in writing within one week of receiving the grade.

A few assignments will allow slightly late submission for a percent deduction on the score. In such cases, the cut-off times and associated penalties will be indicated in the assignment. Assignments submitted beyond the final due date will not be accepted, and you will receive a zero. Not all assignments will allow for late submission. Please read all submission instructions carefully!

This table illustrates the percentage weight of each assessment towards your final grade.

Component		Percentage
Videos		5%
Discussions		15%
Lab Exercises		10%
Quizzes		5%
Assignments		25%
Team Project		
Team Contract (T)	*2%*	
Project Proposal (T)	*5%*	
Project Logical Design (T)	*4%*	
Proposal & Logical Design Review (T)	*2%*	
Progress Report (T)	*5%*	
Peer Reviews (I)	*2%*	
Final Project (T)	*15%*	
Project Status		5%

Letter grades will be assigned using the following scale.

Grading Scale									
A+	≥ 97%	B+	≥ 87.00%	C+	≥ 77.00%	D+	≥ 67.00%		
A	≥ 93.00%	B	≥ 83.00%	C	≥ 73.00%	D	≥ 63.00%	F	<60.0%
A-	≥ 90.00%	B-	≥ 80.00%	C-	≥ 70.00%	D-	≥ 60.00%		

There is no rounding of numeric grades up to the next letter grade.

Course Schedule: (This schedule is for planning purposes and may change. Deadlines are posted in ELMS. Also see ELMS for current information.)

Week/Date	Session	Topics / Videos / Classwork	Reading	Homework	Team Project Assignments
1 1/25 – 1/29	1	Introduction, Syllabus, Course Expectations Database Ethics	Syllabus | ELMS		
	2	MySQL Server & MySQL Workbench Installation	Appendices A/B | ELMS	MySQL Setup	
2 1/30 – 2/5	3	Relational Database Concepts, SELECT from a single table, Team Project Process & Overview	Chapters 1 & 3 | Project Files		
	4	LAB 1: SELECT & MySQL Workbench	Chapter 2	MySQL Setup Complete	Team Formation Survey
3 2/6 – 2/12	5	Relationship types / Joins / UNIONS	Chapter 4		
	6	LAB 2: Multi-table SELECTs			Form teams. Students without a team will be assigned to a team.
4 2/13 – 2/19	13	Normalization, Dependencies, Relationships	Chapter 10	**Assignment 1:** Basic Queries	Work with team to assign team roles & responsibilities
	8	LAB 3: Normalization			Team Contract
5 2/20 – 2/26	7	Review Normalization INSERT / UPDATE / DELETE	Chapter 5	**Assignment 2:** Multi-Table Queries	
	8	LAB 4: INSERT / UPDATE / DELETE			
6 2/27 – 3/5	9	Entity-Relationship Diagrams, Database Creation, and Indexes	Chapters 11		
	10	LAB 5: ERD Teamwork: Project Proposal			Project Proposal
7 3/6 – 3/12	11	Data Types, Functions	Chapters 8 & 9	**Assignment 3:** Normalization	Project Status #1 [Meet with AMP Mentor to discuss Proposal and clarify next steps].
	12	LAB 6: Data Types & Functions			
8 3/13 – 3/19	13	Aggregate Functions, Summary Queries	Chapter 6		
	14	LAB 7: Summaries Teamwork: Project Logical Design			Project Logical Design
9	**SPRING BREAK**				
10 3/27 – 4/2	15	Subqueries, Common Table Expressions	Chapter 7 | ELMS	**Assignment 4:** C*UD	
	16	LAB 8: Subqueries & CTEs			

Week/Date	Session	Topics	Reading	Homework	Team Project Assignments
11 4/3 – 4/9	17	Views, Stored Program Development	Chapters 12 & 13		ProjectStatus#2 [MeetwithAMPMentor to discussprogressandchallenges].
	18	LAB 9: Forward Engineering / Creating/Altering DBs and Tables			Proposal Review
12 4/10 – 4/16	19	Writing Stored Programs, Transactions, Concurrency, & Locking	Chapters 13 & 14	**Assignment 5:** Complex Queries	
	20	Teamwork: Project Work & Progress Report			
13 4/17 – 4/23	21	Stored Procedures, Functions, Triggers, & Events, ETL & Sample Data Strategies	Chapters 15 & 16 | ELMS		
	22	LAB 10: Stored Programs			Project Progress Report
14 4/24 – 4/30	23	Workbench Data Import, Database Backup & Restore	ELMS | Chapter 19		Project Status #3 [Meet with AMP Mentor to get feedback and plan final steps].
	24	LAB 11: Data Import LAB 12: Backup/Restore & Reverse Engineering			
15 5/1 – 5/7	25	Database Administration & Security	Chapters 17 & 18	**Assignment 6:** Views,Functions, & Procedures	
	26	Teamwork: Forward Engineering / Data Strategies & Data Import			
16 5/8 – 5/14	27	Teamwork: SampleData/Backup&Restore/ Queries Course Wrap-up		☐ ProjectDB&FinalReport ☐ Project Reflection	
	28	No Lab Sessions. Thursday, 5/11 is the last day of classes.			

The class meets twice a week for 50 minutes per session unless otherwise noted on page 1 of this syllabus.